A TRUNCATED PERRON FORMULA AND SHORT SUMS OF $\Lambda(n)$ IN ARITHMETIC PROGRESSIONS

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Abstract. We give a self-contained proof of an explicit truncated form of Perron's formula for Dirichlet series $F(s)=\sum_{n\ge 1} a_n n^{-s}$ with $|a_n|\le A(n)$, and use it to estimate $\psi(x;q,a)=\sum_{n\le x,\ n\equiv a \bmod q}\Lambda(n)$ in short intervals $[x,x+h]$ with $h=x^{\theta}$, $\theta>7/12$. The argument is classical; the point is to make every constant explicit.

1. Introduction

Let Λ denote the von Mangoldt function, so that $\Lambda(n)=\log p$ if $n=p^k$ is a prime power and $\Lambda(n)=0$ otherwise, and let

$$\psi(x)=\sum_{n\le x}\Lambda(n)=\sum_{p^k\le x}\log p. \tag{1.1}$$

The Prime Number Theorem in the form $\psi(x)\sim x$ is equivalent to the non-vanishing of the Riemann zeta function on the line $\operatorname{Re} s=1$, and the strongest known error term,

$$\psi(x)=x+O\!\left(x\exp\left(-c\,\frac{(\log x)^{3/5}}{(\log\log x)^{1/5}}\right)\right), \tag{1.2}$$

comes from the Vinogradov–Korobov zero-free region [4, 6]. The link between ψ and the zeros of $\zeta(s)$ is Perron's formula, which recovers the partial sums of a Dirichlet series

$$F(s)=\sum_{n=1}^{\infty}\frac{a_n}{n^s},\qquad \operatorname{Re} s>\sigma_a,$$

from its values on a vertical line. For $F=\zeta$ we have the Euler product and its logarithmic derivative,

$$\zeta(s)=\prod_{p}\bigl(1-p^{-s}\bigr)^{-1},\qquad -\frac{\zeta'}{\zeta}(s)=\sum_{n=1}^{\infty}\frac{\Lambda(n)}{n^s}\quad(\sigma>1). \tag{1.3}$$

Our main results are an explicit truncated Perron formula (theorem 2.3) and its application to primes in short intervals (theorem 3.2). Section 2

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collects the necessary estimates, e.g. the bound $\sum_{n\le x}\Lambda(n)\log\frac{x}{n} = x + O\big(x\exp(-c\sqrt{\log x})\big) + \sum_{p\le x}\frac{\log p}{p}\log\frac{x}{p} - \sum_{p\le\sqrt{x}}\log p$ which we use repeatedly.[1]

Notation. Throughout, p denotes a prime, $s=\sigma+it$ with $\sigma,t\in\mathbb{R}$, and for $x\notin\mathbb{Z}$

$$\sum_{n\le x}{}' a_n = \sum_{n<x} a_n + \tfrac{1}{2}a_x\,\mathbf{1}_{x\in\mathbb{N}}.$$

2. Preliminaries and Perron's formula

2.1. **A discontinuous integral.** For $y>0$ and $c>0$ define the step function

$$\delta(y)=\begin{cases}0 & \text{if } 0<y<1,\\ \tfrac{1}{2} & \text{if } y=1,\\ 1 & \text{if } y>1.\end{cases} \tag{2.1}$$

Lemma 2.1. *Let $c>0$, $y>0$ and $T>0$, and put $I(y,T)=\frac{1}{2\pi i}\int_{c-iT}^{c+iT} y^s\frac{\mathrm{d}s}{s}$. Then*

$$|I(y,T)-\delta(y)| < \begin{cases} y^c\min\left(1,\dfrac{1}{T|\log y|}\right) & \text{if } y\neq 1,\\[2mm] \dfrac{c}{T} & \text{if } y=1.\end{cases} \tag{2.2}$$

Proof. Suppose $y>1$. By Cauchy's theorem applied to the rectangle with vertices $c\pm iT$ and $-U\pm iT$, and letting $U\to\infty$,

$$I(y,T)-1 = -\frac{1}{2\pi i}\left(\int_{-\infty+iT}^{c+iT} - \int_{-\infty-iT}^{c-iT}\right) y^s\frac{\mathrm{d}s}{s}, \tag{2.3}$$

and each horizontal integral is bounded by

$$\left|\int_{-\infty\pm iT}^{c\pm iT} y^s\frac{\mathrm{d}s}{s}\right| \le \frac{1}{T}\int_{-\infty}^{c} y^\sigma\,\mathrm{d}\sigma = \frac{y^c}{T\log y}.$$

For the bound y^c one instead uses the arc of the circle $|s|=|c+iT|$ to the left of the line. The case $0<y<1$ is symmetric, and $y=1$ is a direct computation: $I(1,T)=\frac{1}{\pi}\arctan\frac{T}{c}=\frac{1}{2}-\frac{1}{\pi}\arctan\frac{c}{T}$. □

2.2. **The truncated formula.**

Definition 2.2. A sequence (a_n) is *admissible* if

(i) the series $\sum_n a_n n^{-\sigma}$ converges absolutely for $\sigma>1$, and
(ii) $|a_n|\le A(n)$ for a non-decreasing function A.

[1] All implied constants are absolute unless indicated otherwise; we write $f\ll g$ for $f=O(g)$.

Theorem 2.3 (Truncated Perron formula)**.** *Let (a_n) be admissible, $x \geq 2$, $T \geq 2$, and $c = 1 + \frac{1}{\log x}$. Then*

(2.4)
$$\sum_{n \leq x}{}' a_n = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} F(s) \frac{x^s}{s} \, \mathrm{d}s + O\left(\frac{x}{T} \sum_{n=1}^{\infty} \frac{|a_n|}{n^c} + \frac{xA(2x) \log x}{T} + A(N) \min\left(1, \frac{x}{T\langle x \rangle}\right) \right),$$

where N is the integer nearest to x.

Proof. By absolute convergence (condition (i)) we may integrate term by term:

$$\frac{1}{2\pi i} \int_{c-iT}^{c+iT} F(s) \frac{x^s}{s} \, \mathrm{d}s = \sum_{n=1}^{\infty} a_n \, I\Big(\frac{x}{n}, T\Big).$$

Using lemma 2.1 with $y = x/n$ and splitting the range of n into $n \leq x/2$, $x/2 < n \leq 2x$ and $n > 2x$, we obtain

$$\sum_{n \leq x/2} + \sum_{n > 2x} \ll \frac{x^c}{T} \sum_n \frac{|a_n|}{n^c} \ll \frac{x}{T} \sum_n \frac{|a_n|}{n^c}, \tag{2.5}$$

(2.6)
$$\sum_{x/2 < n \leq 2x} \ll A(2x) \sum_{x/2 < n \leq 2x} \min\left(1, \frac{x}{T|x-n|}\right) \ll \frac{xA(2x)\log x}{T} + A(N) \min\left(1, \frac{x}{T\langle x \rangle}\right),$$

since $x^c = e\,x$ and $|\log(x/n)| \asymp |x-n|/x$ in the middle range; condition (ii) controls $|a_n|$ there. Adding (2.5) and (2.6) gives (2.4). □

Corollary 2.4. *For $2 \leq T \leq x$,*

$$\psi(x) = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} \left(-\frac{\zeta'}{\zeta}(s)\right) \frac{x^s}{s} \, \mathrm{d}s + O\left(\frac{x \log^2 x}{T}\right). \tag{2.7}$$

Proof. Take $a_n = \Lambda(n)$ and $A(n) = \log n$ in theorem 2.3, and use $\sum_n \Lambda(n) n^{-c} = -\frac{\zeta'}{\zeta}(c) \asymp \frac{1}{c-1} = \log x$. □

Remark 2.5. With the smooth weight $\log(x/n)$ the truncation error improves, because

$$\sum_{n \leq x} \Lambda(n) \log \frac{x}{n} = \frac{1}{2\pi i} \int_{(c)} \left(-\frac{\zeta'}{\zeta}(s)\right) \frac{x^s}{s^2} \, \mathrm{d}s$$

converges absolutely; see [1, §17] and [5, Thm. 3.12].

3. Primes in short intervals

We now apply theorem 2.3 with $a_n = \Lambda(n)\chi(n)$, so that $F(s) = -\frac{L'}{L}(s, \chi)$. Write $s = \sigma + it$ throughout, and let $\rho = \beta + i\gamma$ denote a non-trivial zero of $L(s, \chi)$.

Lemma 3.1 (Explicit formula)**.** *Let $2 \le T \le x$ and let χ be a primitive character modulo q. Then*

$$\psi(x,\chi) = \delta_\chi x - \sum_{|\gamma|\le T} \frac{x^\rho}{\rho} - (1-\delta_\chi)\frac{L'}{L}(0,\chi) + O\!\left(\frac{x\log^2(qxT)}{T} + \log x \min\!\Big(1, \frac{x}{T\langle x\rangle}\Big)\right), \tag{3.1}$$

where $\delta_\chi = 1$ if χ is principal and 0 otherwise, and $\langle x\rangle$ denotes the distance from x to the nearest prime power other than x itself.

Proof. Apply (2.4) with $c = 1 + 1/\log x$ and move the line of integration to $\operatorname{Re} s = -\frac{1}{2}$, as in Figure 1. The residues are

$$\operatorname*{Res}_{s=1}\left(-\frac{L'}{L}(s,\chi)\frac{x^s}{s}\right) = \delta_\chi x, \qquad \operatorname*{Res}_{s=\rho}\left(-\frac{L'}{L}(s,\chi)\frac{x^s}{s}\right) = -\frac{x^\rho}{\rho}. \tag{3.2}$$

The horizontal segments contribute $\ll x\log^2(qT)/(T\log x)$ by the standard bound $\frac{L'}{L}(\sigma \pm iT,\chi) \ll \log^2(qT)$, valid for T chosen at distance $\gg 1/\log(qT)$ from every ordinate γ.[2] □

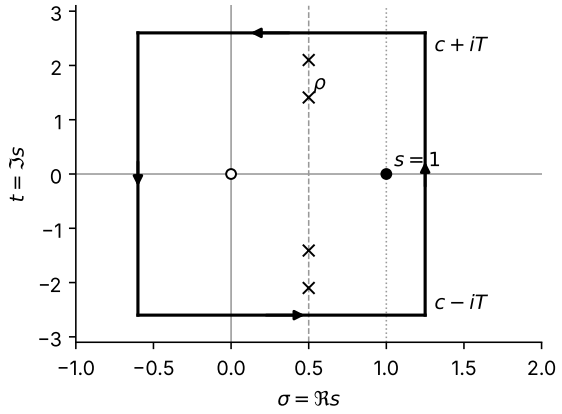


FIGURE 1. The contour used in the proof of lemma 3.1: the segment $[c - iT, c + iT]$ is shifted to the left of the critical line, picking up the pole at $s = 1$ and the zeros ρ with $|\gamma| \le T$.

Subtracting (3.1) at x and $x + h$ and averaging over characters gives the following.

[2] Such a T exists in every interval $[T_0, T_0 + 1]$, since there are $O(\log qT_0)$ zeros with $|\gamma - T_0| \le 1$.

Theorem 3.2. *Let $\varepsilon > 0$ and $\theta > \frac{7}{12} + \varepsilon$. For $x \geq x_0(\varepsilon)$, $h = x^\theta$, and $q \leq (\log x)^A$,*

$$\psi(x+h;q,a) - \psi(x;q,a) = \frac{h}{\varphi(q)}\left(1 + O\left(\exp\left(-c\frac{\log x}{(\log\log x)^{2/3}}\right)\right)\right) \tag{3.3}$$

uniformly for $(a,q) = 1$.

Proof. By orthogonality of characters,

(3.4)

$$\begin{aligned}\psi(x+h;q,a) - \psi(x;q,a) &= \frac{1}{\varphi(q)} \sum_{\chi \bmod q} \bar\chi(a)\big(\psi(x+h,\chi) - \psi(x,\chi)\big) \\ &= \frac{h}{\varphi(q)} - \frac{1}{\varphi(q)} \sum_{\chi \bmod q} \bar\chi(a) \sum_{|\gamma|\leq T} \frac{(x+h)^\rho - x^\rho}{\rho} + O\left(\frac{x\log^2 x}{T}\right).\end{aligned}$$

Since $|(x+h)^\rho - x^\rho| \leq h x^{\beta-1}|\rho|$, the double sum is at most

$$h \sum_{\chi} \sum_{|\gamma|\leq T} x^{\beta-1} \leq h \int_0^1 x^{\sigma-1}\,\mathrm{d}N(\sigma,T;q) \ll h\,(\log x) \max_{1/2\leq\sigma\leq 1} x^{\sigma-1} N(\sigma,T;q),$$

where $N(\sigma,T;q)$ counts zeros of all $L(s,\chi)$, $\chi \bmod q$, with $\beta \geq \sigma$ and $|\gamma| \leq T$. Huxley's density estimate $N(\sigma,T;q) \ll (qT)^{\frac{12}{5}(1-\sigma)}(\log qT)^B$ [3, Ch. 12] and the Vinogradov–Korobov zero-free region [4, 6] finish the proof with $T = x^{1-\theta+\varepsilon}$. □

Remark 3.3. The exponent $\frac{7}{12}$ is Huxley's [2]; any improvement in the density hypothesis range transfers directly. The matrix

$$\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}), \qquad \gamma \cdot z = \frac{az+b}{cz+d},$$

plays no role here but is a reminder that the same method applies to Hecke L-functions, for which the relevant maps fit in the diagram

$$\begin{array}{ccc} \mathcal{S}_k(\Gamma_0(N)) & \xrightarrow{\ \Lambda\ } & \mathcal{D} \\ \big\downarrow & & \big\downarrow \mathcal{M} \\ \mathcal{S}_k(\Gamma_1(N)) & \longrightarrow & \mathcal{H} \end{array}$$

of Mellin transforms.

TABLE 1. Admissible exponents θ for theorem 3.2 under various zero-density hypotheses.

Density estimate	exponent $A(\sigma)$	admissible θ
Ingham (1937)	$\frac{3}{2-\sigma}$	$\theta > \frac{5}{8}$
Huxley (1972)	$\frac{12}{5}$	$\theta > \frac{7}{12}$
Density Hypothesis	2	$\theta > \frac{1}{2}$

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